## SoccerGMR SI-Bind: Detailed Component Ablation Study (E1 to E9)

Fine-grained ablation across mechanisms: Mechanism A (Binding & Residual), Mechanism B (Slot Innovation), and Mechanism C (Dominance Refresh) on Standard Test (1,036 queries) and Full Test (2,970 queries).

| | | Standard Test Split (1,036 queries) | | | | Full-Decodable Test Split (2,970 queries) | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Backbone** | **Ablation Method / Configuration** | **mAP (%)** | **Orc-Gap (%)** | **Rej-F1@.5 (%)** | **AUROC (%)** | **mAP (%)** | **Orc-Gap (%)** | **Rej-F1@.5 (%)** | **AUROC (%)** |
| **Moment-DETR** | 1. E1: Binding-only (Interlayer Anchor) | 12.32 | **18.82** | 75.18 | **75.96** | 10.34 | **17.98** | 78.49 | **82.38** |
| | 2. E2: Lite Binding + Residual Stream | 11.85 | 22.41 | 72.40 | 71.55 | 10.14 | 21.20 | 76.11 | 80.14 |
| | **3. E3: E2 + Slot-Private Innovation (B)** | **15.60** | 24.26 | 66.22 | 73.17 | **13.30** | 23.02 | 67.12 | 80.91 |
| | 4. E4: E2 + Uniform Random Refresh (C Control) | 12.35 | 24.02 | 60.18 | 75.53 | 12.13 | 22.83 | 61.91 | 81.77 |
| | **5. E5: E2 + Dominance-Aware Refresh (C)** | 7.95 | 21.56 | 75.00 | 68.32 | 6.47 | 21.06 | **78.77** | 77.80 |
| | **6. E6: E2 + Innovation (B) + Dominance (C)** | 15.21 | 23.66 | 65.52 | 74.32 | 12.58 | 22.21 | 68.83 | 81.86 |
| | 7. E7: E6 - Binding (bind_coef=0 Ablation) | 15.21 | 23.66 | 65.52 | 74.32 | 12.58 | 22.21 | 68.83 | 81.86 |
| | **8. E8: Strict Serial A -> B (No Refresh C)** | 12.44 | 22.03 | **75.57** | 73.71 | 10.51 | 20.95 | 78.37 | 81.21 |
| | **9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)** | 13.35 | 21.60 | 63.61 | 74.93 | 12.99 | 19.56 | 66.27 | 79.12 |
| **BM-DETR** | 2. E2: Lite Binding + Residual Stream | 6.36 | 18.84 | **75.00** | 68.63 | 5.68 | 18.46 | 78.80 | 78.79 |
| | **3. E3: E2 + Slot-Private Innovation (B)** | 7.88 | 21.63 | 72.74 | 71.55 | 7.08 | 20.95 | 76.91 | 79.45 |
| | 4. E4: E2 + Uniform Random Refresh (C Control) | 6.48 | 20.60 | **75.00** | 69.37 | 5.95 | 19.98 | **78.91** | 79.18 |
| | **5. E5: E2 + Dominance-Aware Refresh (C)** | 7.16 | **15.26** | **75.00** | 69.37 | 6.93 | **15.16** | 78.14 | 78.92 |
| | **6. E6: E2 + Innovation (B) + Dominance (C)** | 7.72 | 20.91 | 73.40 | 70.18 | 7.03 | 20.30 | 77.47 | 79.27 |
| | 7. E7: E6 - Binding (bind_coef=0 Ablation) | 7.72 | 20.91 | 73.40 | 70.18 | 7.03 | 20.30 | 77.47 | 79.27 |
| | **8. E8: Strict Serial A -> B (No Refresh C)** | **12.26** | 21.71 | 70.99 | **73.09** | **10.92** | 20.48 | 76.14 | **80.21** |
| | **9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)** | 10.72 | 23.09 | 70.85 | 72.73 | 9.54 | 21.78 | 75.34 | 80.12 |
| **UVCOM** | 1. E1: Binding-only (Interlayer Anchor) | 15.31 | 18.42 | 74.94 | 77.68 | 14.60 | 17.55 | 73.74 | 78.58 |
| | 2. E2: Lite Binding + Residual Stream | 20.31 | **14.46** | 72.79 | 77.04 | 20.02 | **14.54** | 70.95 | 76.70 |
| | **3. E3: E2 + Slot-Private Innovation (B)** | **20.87** | 18.11 | 69.84 | 77.23 | **20.95** | 16.99 | 69.72 | 76.14 |
| | 4. E4: E2 + Uniform Random Refresh (C Control) | 17.30 | 16.39 | 74.17 | 77.79 | 16.34 | 15.66 | **73.78** | **78.94** |
| | **5. E5: E2 + Dominance-Aware Refresh (C)** | 20.49 | 17.55 | 69.48 | **78.13** | 19.76 | 16.74 | 68.12 | 75.69 |
| | **6. E6: E2 + Innovation (B) + Dominance (C)** | 18.99 | 16.30 | **75.40** | 76.91 | 17.74 | 15.55 | 73.11 | 75.20 |
| | 7. E7: E6 - Binding (bind_coef=0 Ablation) | 18.99 | 16.30 | **75.40** | 76.91 | 17.74 | 15.55 | 73.11 | 75.20 |
| | **8. E8: Strict Serial A -> B (No Refresh C)** | 11.64 | 22.85 | 71.00 | 72.64 | 10.77 | 21.95 | 71.33 | 76.12 |
| | **9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)** | 19.74 | 17.28 | 72.89 | 77.84 | 18.58 | 16.69 | 69.85 | 75.33 |
| **CG-DETR** | 2. E2: Lite Binding + Residual Stream | 14.99 | 23.42 | 72.77 | 73.86 | 12.68 | 22.41 | 75.71 | **81.67** |
| | **3. E3: E2 + Slot-Private Innovation (B)** | 16.29 | 24.97 | 64.69 | 75.17 | 14.54 | 23.68 | 65.94 | 80.57 |
| | 4. E4: E2 + Uniform Random Refresh (C Control) | 14.47 | **20.87** | 75.19 | 74.40 | 13.00 | **19.67** | **78.33** | 81.20 |
| | **5. E5: E2 + Dominance-Aware Refresh (C)** | 13.80 | 21.77 | **75.72** | 75.42 | 13.96 | 21.14 | 77.43 | 81.33 |
| | **6. E6: E2 + Innovation (B) + Dominance (C)** | 17.38 | 22.89 | 70.85 | 75.92 | 15.54 | 21.61 | 73.14 | 80.94 |
| | 7. E7: E6 - Binding (bind_coef=0 Ablation) | 17.38 | 22.89 | 70.85 | 75.92 | 15.54 | 21.61 | 73.14 | 80.94 |
| | **8. E8: Strict Serial A -> B (No Refresh C)** | 15.22 | 23.40 | 71.42 | 74.25 | 13.36 | 22.11 | 73.67 | 81.33 |
| | **9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)** | **18.76** | 21.55 | 69.27 | **77.50** | **17.41** | 20.21 | 70.53 | 81.24 |
| **BAM-DETR** | 2. E2: Lite Binding + Residual Stream | 22.57 | **19.78** | **71.79** | 78.04 | 20.75 | **18.73** | **72.80** | **79.91** |
| | **3. E3: E2 + Slot-Private Innovation (B)** | 22.21 | 22.18 | 68.50 | 77.30 | 20.34 | 21.06 | 67.36 | 77.80 |
| | **5. E5: E2 + Dominance-Aware Refresh (C)** | 23.52 | 21.13 | 67.14 | 78.18 | 20.72 | 19.33 | 67.02 | 77.81 |
| | **6. E6: E2 + Innovation (B) + Dominance (C)** | 23.78 | 20.40 | 68.96 | 78.37 | 21.78 | 19.07 | 69.28 | 78.69 |
| | **8. E8: Strict Serial A -> B (No Refresh C)** | 22.47 | 22.63 | 68.11 | 78.67 | 21.08 | 21.08 | 67.47 | 78.65 |
| | **9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)** | **24.21** | 20.68 | 69.53 | **78.96** | **22.17** | 19.15 | 69.12 | 79.03 |
| **Sim-DETR** | 2. E2: Lite Binding + Residual Stream | 18.99 | 21.89 | 75.11 | 73.55 | 16.94 | 21.13 | **77.25** | **79.91** |
| | **3. E3: E2 + Slot-Private Innovation (B)** | 20.95 | 18.92 | **76.07** | **75.34** | 19.58 | 18.64 | 75.29 | 78.62 |
| | **5. E5: E2 + Dominance-Aware Refresh (C)** | 19.14 | 21.15 | 75.00 | 72.22 | 17.31 | 20.43 | 75.04 | 78.05 |
| | **6. E6: E2 + Innovation (B) + Dominance (C)** | 21.17 | 21.26 | 74.89 | 75.12 | 20.08 | 20.83 | 73.10 | 77.83 |
| | **8. E8: Strict Serial A -> B (No Refresh C)** | 15.69 | **16.83** | 75.00 | 71.80 | 14.54 | **17.04** | 74.23 | 78.70 |
| | **9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)** | **21.34** | 23.16 | 72.12 | 74.70 | **20.46** | 22.30 | 73.46 | 79.47 |
| **EaTR** | 2. E2: Lite Binding + Residual Stream | 16.32 | 21.30 | **74.58** | 75.27 | **15.43** | 20.24 | **75.47** | 79.21 |
| | **3. E3: E2 + Slot-Private Innovation (B)** | 15.11 | 23.84 | 61.97 | 76.76 | 14.73 | 22.33 | 64.05 | 82.50 |
| | **5. E5: E2 + Dominance-Aware Refresh (C)** | **17.56** | 24.33 | 65.09 | 76.10 | 14.96 | 22.85 | 66.40 | 82.30 |
| | **6. E6: E2 + Innovation (B) + Dominance (C)** | 17.12 | 21.80 | 67.74 | 76.23 | 15.35 | 20.89 | 68.66 | 81.03 |
| | **8. E8: Strict Serial A -> B (No Refresh C)** | 6.27 | **21.04** | 74.33 | 70.50 | 5.34 | 21.12 | 73.68 | 78.74 |
| | **9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)** | 17.17 | 21.58 | 69.33 | **77.21** | 15.25 | **19.81** | 72.41 | **82.55** |
| **QD-DETR** | 2. E2: Lite Binding + Residual Stream | 16.37 | 24.11 | 60.54 | **76.84** | 13.06 | 22.36 | 63.51 | **83.60** |
| | **3. E3: E2 + Slot-Private Innovation (B)** | 12.71 | 22.15 | 72.82 | 74.74 | 10.70 | 21.36 | 74.96 | 82.04 |
| | **5. E5: E2 + Dominance-Aware Refresh (C)** | 15.21 | 20.98 | 72.08 | 75.61 | 12.17 | **19.67** | 74.36 | 82.21 |
| | **6. E6: E2 + Innovation (B) + Dominance (C)** | 16.30 | **20.65** | **73.71** | 76.63 | 13.20 | 19.84 | **75.91** | 82.47 |
| | **8. E8: Strict Serial A -> B (No Refresh C)** | 14.53 | 23.75 | 64.46 | 75.85 | 12.93 | 22.37 | 66.69 | 83.00 |
| | **9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)** | **16.77** | 24.35 | 59.41 | 75.27 | **14.96** | 22.68 | 61.60 | 81.78 |

Note: E1 = Binding only (identity across layers); E2 = Lite Binding + Residual; E3 = E2 + Slot-Private Innovation (B); E4 = E2 + Uniform Random Refresh (C Control); E5 = E2 + Dominance Refresh (C); E6 = E2+B+C (Champion Triad); E7 = E6 - Binding; E8 = Strict Serial A -> B without C; E9 = Strict Serial C -> A -> B (Refresh -> Re-bound -> Nov).
Bold numbers indicate best performance among ablation variants within the same backbone. Oracle Gap: lower is better.